

Survey on Artificial Intelligence Tool Usage for Assignment Preparation

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Survey on Artificial Intelligence Tool Usage for Assignment Preparation

A quantitative study investigating AI tool adoption, usage patterns, and student perceptions at the Sri Lanka Institute of Information Technology (SLIIT). Submitted to the Faculty of Computing, SLIIT

May 1, 2026

TERMS OF REFERENCE

This report is submitted in partial fulfillment of the conditions for the module IT 1150. The ideal of this report is to dissect and bandy the data collected regarding the operation of Artificial Intelligence(AI) tools for assignment medication among scholars at the SLIIT. The report is to be completed and submitted by 01.05.2026.

ACKNOWLEDGEMENT

We offer our deepest thanks and appreciation to our mentor, for offering us the essential direction, advice, and constant assistance during the entire course of this project. We are also grateful to the leaders and staff at the SLIIT for making the facilities available to finish this study.

Additionally, we are truly thankful to every SLIIT peer who chose to join our voluntary research poll. Their precious hours and candid input provided the basis for statistical reviews and allowed this paper to happen. Lastly, we thank our teammates for their commitment, unity, and shared hard work in completing this document successfully.

Abstract

The quick adoption of AI technology in academia has greatly changed the way students study. The objective of this quantitative research is to explore the use of AI technologies for assignment preparation by SLIIT students. Data was gathered using an online survey distributed among faculties.

The results show the extensive use of ChatGPT and Gemini, mainly in the context of conducting research, summarizing, and writing. Although many students think that using AI technologies can help them save time and learn better, there is an equally significant number who recognize the danger of obtaining inaccurate and even “hallucinated” knowledge.

Introduction

The development of user-friendly AI platforms has led to a major transformation in the field of education. Software solutions like ChatGPT, Gemini, and GitHub Copilot have been gaining more acceptance as essential instruments for use by university students while undertaking their assignments. These platforms provide unparalleled functionality in the production of content, analysis of data, and programming. Yet, the degree of dependence that university students may have on these platforms, their utility at different phases of assignment preparation, and advantages or disadvantages have been topics of debate in academia.

This research focuses on students at the SLIIT and seeks to understand their specific behaviors regarding AI tool usage. The primary objectives of this study are:

Tool Popularity

Identify the most popular AI tools utilized by SLIIT students for academic purposes.

Frequency & Application

Determine the frequency and specific applications of AI tools during assignment preparation.

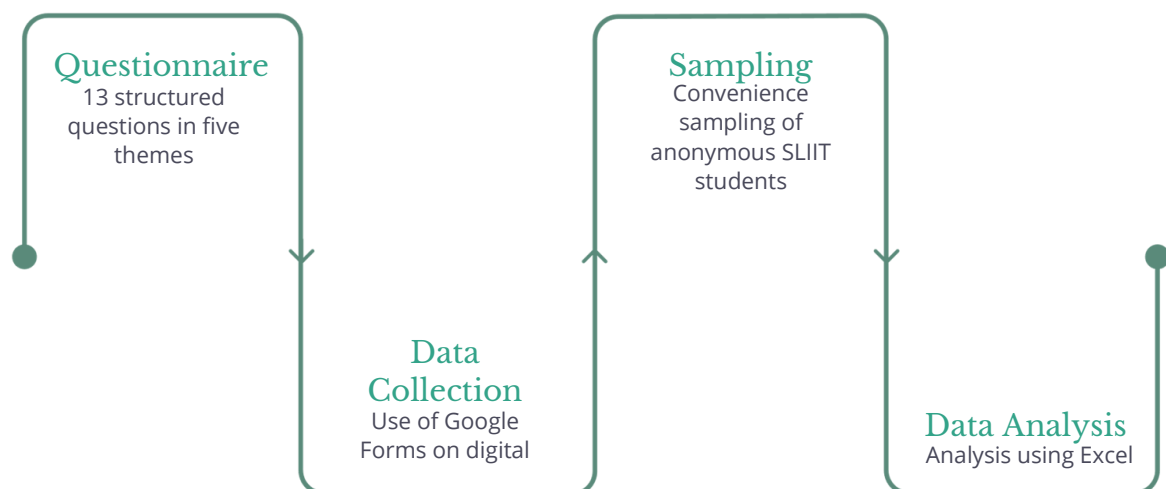
Perceptions

Evaluate students' perceptions regarding helpfulness and awareness of potential AI hallucinations.

After considering all these issues, this paper is going to present a full picture of how AI is being integrated into student tasks at the moment, thus helping both students and teachers make optimal use of these technologies without compromising their credibility.

METHODOLOGY

The study utilizes the quantitative research design of surveys to collect numerical and measurable findings from student behaviors. The research process involves several steps:



Data Analysis & Discussion

The following part provides the results obtained from analyzing the survey responses. The results have been organized into different categories based on the main sections in the questionnaire; these include Demographics, Awareness and Use, Purpose and Application, and Perception and Effectiveness.

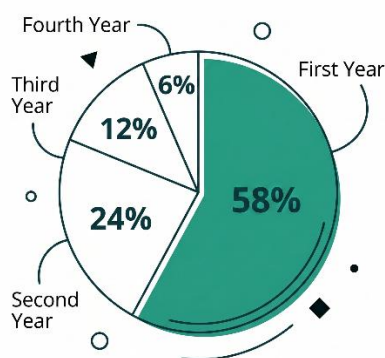
- **Demographics**

The demographic profile of the respondents is important to consider as far as putting the data into context. Two major demographic factors were obtained from the survey, namely, the respondents' level of study and their faculty. Based on the bar graphs below, it can be seen that the majority of the respondents came from the first year and belonged to the Faculty of Information Technology and Computer Science.

Academic Level

First-Years represent a substantial proportion of the sample, implying that new entrants into the universities have been particularly active in embracing digital technology to handle university work.

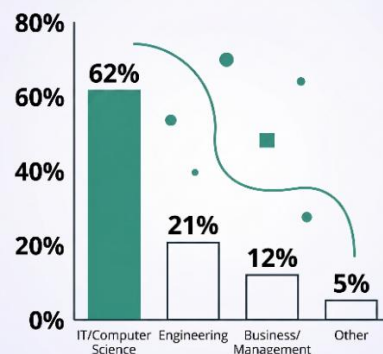
Q01 Academic Level (Pie Chart)



Primary Faculty

The dominant faculties among respondents include Information Technology/Computer Science, Engineering, and Business/Management, indicating the technical bias of AI pioneers.

Q02 Faculty (Bar Chart)



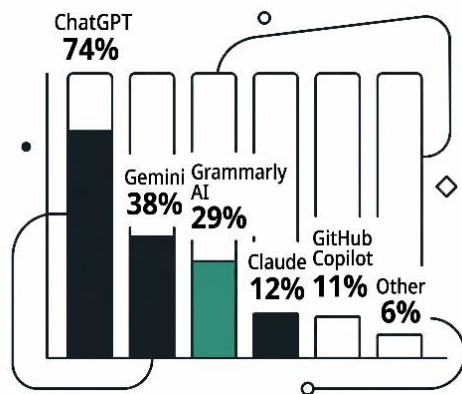
First-year students constitute the majority of respondents (58%), implying that recent arrivals use technology to adjust to university standards. The large percentage within Information Technology/Computer Science (62%) and Engineering (21%) is justified by the nature of these courses, which embrace new technologies faster than others.

- **Awareness & Usage Patterns**

This sub-section examines what technologies students use and how often they integrate these into their weekly schedule. It is clear from the data that AI has made the leap from being merely novel to a central, consistent feature of students' academic routine at SLIIT.

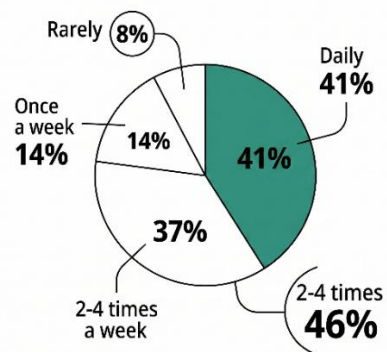
Most Frequently Used AI Tools

The AI tool that was found to be used by most SLIIT students was ChatGPT. Gemini and Grammarly (AI writing capabilities) are used to a considerable extent too. This preference towards these AI tools reflects the heavy dependency on conversational AI models.



Weekly Usage Frequency

The overwhelming majority of students make use of AI technologies either "Daily" or "2-4 times per week," while only a few of them use such technologies "Rarely (only for big assignments)," showing that AI is now well integrated into the studying routine.



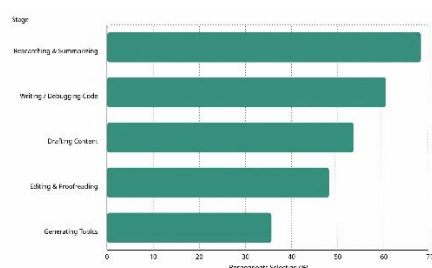
ChatGPT leads by far, accounting for 74% of total AI usage, while Gemini is used at 38% and Grammarly's AI capabilities are used at 29%. The most important statistic, however, is that 78% of the students make use of AI technologies on a daily or weekly basis.

- **Purpose & Application**

This knowledge can give us information about the specific academic needs that artificial intelligence meets. The results reveal that students rely on AI not only at the end of the writing process, but also during the stage of thinking at the beginning of research.

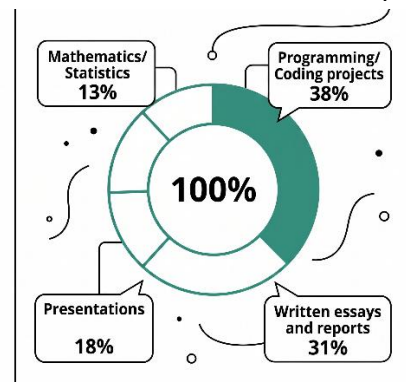
Stages of Assignment Preparation

The tasks that are the most often selected include "Researching and summarizing long articles" and "Writing or debugging code" (not surprising, given the predominance of the IT students). "Creating drafts of text" and "Proofreading/editing" are also common.



Most Likely to Use AI

It can be seen that there is a heavy dependence on AI when it comes to "Programming / Coding projects" and "Written essays and reports." This is due to the abilities offered by sophisticated tools such as ChatGPT and GitHub Copilot.



Students are not only employing AI technologies for post-editing purposes but are heavily relying on them at the very beginning stage of cognition through research and concept formulation. The high importance attributed to code writing help (61%) and programming projects (38%) is clearly indicative of the IT-professional respondent sample size.

- Perceptions & Effectiveness

4.2

Average Helpfulness Rating

On a 1–5 scale (Q08), reflecting overwhelmingly positive student perception of AI's value in improving assignment quality.

77%

Rate AI as 4 or 5

The vast majority gave top-tier helpfulness scores, underscoring the practical value students derive from these technologies.

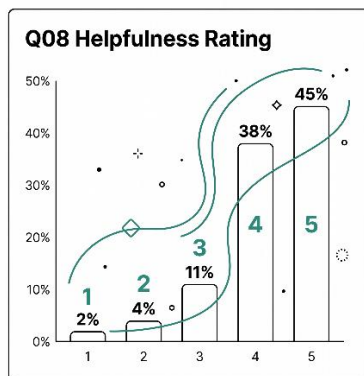
63%

Encountered Hallucinations

Students who answered "Yes" or "Maybe" to Q10, indicating awareness that AI is not infallible and can produce factually incorrect outputs.

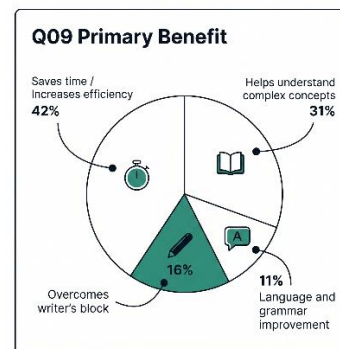
Helpfulness Rating (1–5 Scale)

The vast majority of participants assessed the usefulness of the AI tools to be a rating of either 4 or 5. Such an overwhelmingly positive attitude emphasizes the benefits that students gain from such technologies in improving their academic performance.



Primary Benefit Gained

"Time saving/ increases efficiency" and "Makes it easier to grasp complex concepts" are the two most common advantages. This clearly indicates that AI not only increases efficiency but also plays the role of a personal tutor for complex subjects".



AI serves both as an amplifier of efficiency and a personal tutor on challenging topics. Most importantly, 63% having experienced hallucination proves that the students are not completely blind followers of the machine's results – a minimum of critical thinking is still utilized to evaluate them.

- **Ethical Considerations and Academic Integrity**

With the emergence of artificial intelligence, academic integrity has become an issue of paramount importance to educational institutions. This survey investigated the level of knowledge that students have about institutional policies on plagiarism.

Aware of your university's official policy regarding the use of AI tools in coursework

The findings have shown that there is a mixture of levels of awareness within the student community. While some students marked "Yes" showing that they know about the official position of SLIIT, others marked "No" or "Maybe." It becomes evident that an essential communication breakdown exists since students are using platforms such as ChatGPT and Gemini extensively but without knowing the academic limitations imposed by the institution.

Concerned about committing unintentional plagiarism when using AI tools

Curiously, opinions on this topic range widely. A large number of students gave ratings of 4 or 5, indicating a fairly high level of concern regarding accidental plagiarism. In other words, while the students see the usefulness of AI technology, they also feel uneasy due to its ethical consequences and potential flagging of their paper by anti-AI software. Alternatively, those students who scored themselves low on the scale may be knowledgeable enough about proper synthesis of AI thoughts, or they simply do not know what AI-based plagiarism is.

Psychological Impact and Stress Reduction

The qualitative data analyzed yielded strong emergence of themes like "smart planning & time management." The students said that artificial intelligence serves as a planning tool by assisting them in segmenting difficult assignments into smaller components. In addition, the respondents mentioned that using artificial intelligence to help understand complicated concepts helps to lower the level of anxiety regarding difficult subjects. As a result, artificial intelligence works as an online tutor, making them feel less "stuck," thus lowering their stress level at university."

Conclusion

This study aimed to examine the usage of AI-based tools in assignment writing among SLIIT students. From the quantitative results that have been obtained, a number of important findings can be observed. First, the usage rate of AI-based tools is very high, especially among first-year students and those enrolled in IT/Engineering majors, and ChatGPT has emerged as the most favored tool by far.

Secondly, students are not only using AI-based tools for simple grammatical corrections but are fully integrating them into their work process, using them extensively for research, summarization, writing, and programming.

In conclusion, while students consider these resources to be extremely useful in saving time and understanding complicated subjects (giving them very high ratings for usefulness), they are not completely blind followers. A vast majority of students have realized that AI may provide inaccurate or "hallucinated" answers. This implies that while students use AI to make things easier, some degree of critical thinking is employed to ensure that the results are accurate. In summary, AI tools have undoubtedly been a valuable resource for SLIIT students, revolutionizing the entire process of preparing assignments.

Recommendation

Based on the results and findings from this survey, the following recommendations can be made:

- **Fact-Checking Guidelines**

Given that many students have received "hallucinated" data, it is imperative that students become accustomed to double-checking AI-produced facts, citations, and code logic against reputable academic literature or textbooks.

- **Prompt Engineering Workshops**

It is advisable that academic institutions offer courses on "Prompt Engineering." It will help students to craft better questions and receive better answers from these AI programs.

- **Promote AI as a Tutor, Not a Writer**

Students should be advised to use these tools for the sole purpose mentioned in the survey—understanding complex concepts—rather than using them to produce their essays, in order to retain their own academic integrity.

- **Clear Institutional Policies**

Given that most students utilize these tools regularly, it is important that academic institutions issue clear guidelines for when and how to utilize such tools.

References

[1] Monash University, "IT report structure," Monash University Research and Learning Online. [Online]. Available: <http://www.monash.edu/rlo/assignment-samples/information-technology/it-report-structure>. [Accessed: 28-Apr-2026]

Appendices

Survey Questionnaire: Survey on Artificial Intelligence Tool Usage for Assignment Preparation

- 01.** What is your current academic level?
- 02.** What is your primary faculty or field of study?
- 03.** Which of the following AI tools do you use most frequently for academic purposes?
- 08.** On a scale of 1 to 5, how helpful do you find AI tools in improving the overall quality of your assignments? (1 is low, 5 is high)
- 09.** What is the primary benefit you gain from using AI tools?
- 10.** Have you ever noticed AI tools giving you incorrect or "hallucinated" information for an assignment?
- 11.** Are you aware of your university's official policy regarding the use of AI tools in coursework?
- 12.** How concerned are you about committing unintentional plagiarism when using AI tools? (1 is low, 5 is high)
- 13.** How do you decrease stress with the use of AI Tools?